

Milestone-F Test Automation

Six SauceDemo workflows in the maintained Selenium lane

1 Document details

Field	Details
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Work period	26 September 2026 to 3 October 2026 (UTC+08:00)
Repository	selenium-saucedemo-remediation
Application	SauceDemo https://www.saucedemo.com
Final passing run	20260930-210141-857 six F tests passed
Document date	3 October 2026

2 Purpose and outcome

Milestone-F uses SauceDemo to show current Selenium work in an inherited Java repository. I developed six Java TestNG tests in the maintainedTest lane established in Milestone-B through Milestone-D, using manual cases completed in Milestone-E. The verified run on 30 September 2026 passed all six tests. Gradle returned exit code 0, confirming that the command finished successfully. The following sections explain how the tests were implemented and how their results were verified.

3 Work completed

Activity	Work performed	Outcome
M-F-A1	Recorded the baseline, workbook hashes, Milestone E-to-F links and Chrome control observations.	Chrome readiness passed in the maintained lane. The case map and boundaries were documented.
M-F-A2	Developed the dedicated Gradle task and six TestNG methods. Page and support classes handle the browser interactions.	Six workflows compiled and ran without the inherited Excel, retry or reporting helpers.
M-F-A3	Temporarily changed expected values to exercise failure capture, then restored the source.	The six induced failures made Gradle return exit 1. Failed entries appeared in the HTML/XML reports, with browser images captured before cleanup.
M-F-A4	Ran all six methods with maximized Chrome and reviewed ten checkpoint images.	Final passing run 20260930-210141-857 passed six of six. The Gradle exit code was 0.
M-F-A5	Checked each Milestone-F test against its corresponding Milestone-E case and expected result.	The verified result matched the tested source. I reviewed the ten checkpoint screenshots from the passing run against the expected page states.
M-F-A6	Prepared the Milestone-F PDF and README. The supporting evidence was assembled for publication.	The technical commit was pushed. The Milestone-F PDF was added to the repository, where its README points to the supporting evidence.

4 Workflow coverage and Milestone-E links

Milestone-F uses six independent workflows built around the expected SauceDemo shopping flow. Together, they follow a user from sign-in and product browsing through cart changes to checkout and order completion. Price sorting is a common browsing action. The locked-out login and blank First Name checks show visible points where the flow cannot continue. Keeping the automation close to this familiar journey made the scope manageable without scripting every Milestone-E variation. This lean approach kept the work focused on developing and verifying the six workflows without expanding it into full-site automation. I used these workflows to demonstrate Selenium script development within that scope, without setting employer coverage priorities or conducting a formal risk ranking. Each workflow has an expected outcome from cases already executed in Milestone-E. AUT-F03 and AUT-F06 link related results where the interaction continues across pages. The remaining variations remain manual.

Automation	Manual result used	What the script checks
AUT-F01	TC01 / RES-001	Standard login reaches Products with visible inventory.
AUT-F02	TC02 / RES-002	Locked-out login shows the rejection and stays on sign-in.
AUT-F03	TC05 / RES-008 TC06 / RES-009	Two products enter Cart. Removing Backpack there and Bike Light on Products leaves Cart empty.
AUT-F04	TC07 / RES-010	Blank First Name triggers the required-field error on checkout information.
AUT-F05	TC04 / RES-006	The low-to-high option is selected and every next displayed price is no lower.
AUT-F06	TC08 / RES-014 and RES-016	Two items and totals match on Overview. Finish reaches completion.

5 Implementation in the maintained lane

Milestone-B through Milestone-D established a separate `maintainedTest` source set and used Java 21 for the maintained lane. They also set up strict TestNG execution and a Chrome readiness check for the login page. Milestone-F used this foundation for its functional scripts. In `build.gradle`, the new `milestoneFAutomation` task compiles the maintained source set and runs only the three Milestone-F test classes. The task fails if no tests match its filter or if an executed assertion fails. The final Gradle JUnit-style XML results confirm that all six intended methods ran. Each method starts a fresh `ChromeDriver` and runs once. Test cleanup closes Chrome afterward. Selenium Manager resolves the driver as it did in Milestone-D.

Location in the repository	Contribution in Milestone-F
<code>src/maintainedTest/java/saucedemo/pages/</code>	<code>LoginPage</code> , <code>InventoryPage</code> , <code>CartPage</code> and <code>CheckoutPage</code> hold page actions and locators. Most controls use observed data-test CSS selectors. <code>CheckoutPage</code> uses one text-based XPath for the visible PDF control.
<code>src/maintainedTest/java/saucedemo/support/</code>	<code>TestData</code> keeps demo accounts and synthetic checkout values. <code>BaseUiTest</code> opens Chrome, logs checkpoints, saves screenshots and closes the browser.
<code>src/maintainedTest/java/saucedemo/tests/</code>	<code>AuthenticationTests</code> , <code>InventoryAndCartTests</code> and <code>CheckoutTests</code> contain six TestNG methods and the result assertions.
<code>gradle/timestamped-build.init.gradle</code>	Creates a separate ignored <code>build/timestamped-runs/<run ID>/</code> directory when passed to Gradle. It does not add a test framework.

Milestone-F reused Selenium Java 4.49.0 and TestNG 7.12.0 from the maintained lane. The scripts import `WebDriver`, `By`, `WebDriverWait`, `ChromeDriver` and `TakesScreenshot` from Selenium.

TestNG supplies `@Test` and assertions. Its method hooks handle browser setup and cleanup. `BigDecimal` checks prices and totals, while `java.nio.file` saves PNGs. Gradle provides the console and HTML/XML reports. No new reporting or logging library was added.

The inherited `AutomationPractice` configuration still declares Selenium 3 and TestNG 7.3.0. Its other dependencies remain in place. Milestone-F did not import its Excel/POI data path, Cucumber runner, retry listener, Allure or Logback reporting hooks, Slack client or `WebDriverManager` helper. The default legacy test task suppresses failures. The six Milestone-F tests were run through `milestoneFAutomation`, which fails the Gradle task if a test fails. The older scripts and protected workbooks were left unchanged. The maintained-lane updates let the six Milestone-F tests run in the inherited repository. Their passing results cover the SauceDemo workflows. They do not establish an outcome for the original `AutomationPractice` suite.

6 Execution results and evidence

Run 20260930-210141-857 used Gradle 8.14.5 and Java 21 on Windows 11. The same forced invocation ran the Milestone-D-style readiness check once and the dedicated F task once. The readiness check passed 1 of 1. The Milestone-F task passed all six tests, with no failures or skips. Gradle printed BUILD SUCCESSFUL and the captured exit code was 0. It reported Chrome 154.0.8037.58 and ChromeDriver 154.0.8037.92. The Gradle JUnit-style XML gives the method-level status. The figures below show the generated Gradle summary and checkout class detail.

Test Summary

6 tests	0 failures	0 ignored	10.631s duration	100% successful
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Packages	Classes	Tests	Failures	Ignored	Duration	Success rate
saucedemo.tests		6	0	0	10.631s	100%

Generated by Gradle 8.14.5 at Sep 30, 2026, 9:02:35 PM

Figure 1 Verified F suite summary from run 20260930-210141-857. Six tests passed.

Class saucedemo.tests.CheckoutTests

all > saucedemo.tests > CheckoutTests

2 tests	0 failures	0 ignored	5.646s duration	100% successful
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Tests	Standard output	Standard error
Test	Duration	Result
autF04_firstNamelsRequired	1.722s	passed
autF06_twoltemOrderCompletes	3.924s	passed

Wrap lines ☐
Generated by Gradle 8.14.5 at Sep 30, 2026, 9:02:35 PM

Figure 2 Checkout class detail for AUT-F04 and AUT-F06 from the same run.

6.1 Failure reporting verification

M-F-A3 used temporary wrong expectations to check failure reporting. Run 20260930-122320-911 failed 6 of 6, exited 1 and saved failure PNGs before browser close. The source was restored before the final passing run. This was a diagnostic, not a SauceDemo defect. An earlier AUT-F03 timeout followed a Chrome password warning. AUT-F03 passed after the automation browser preference was set.

6.2 Run output locations

Gradle saved the original run outputs under build/timestamped-runs/ in the local repository checkout. The table below gives paths within those generated folders. Section 6.3 gives the locations of the copies provided under Milestone-F/. The passing run folder is 20260930-210141-857. The diagnostic run folder is 20260930-122320-911.

Evidence in generated run	Path from that run folder
Automation test summary (HTML)	reports/tests/milestoneFAutomation/index.html
Automation test results (JUnit XML)	test-results/milestoneFAutomation/TEST-*.xml

Evidence in generated run	Path from that run folder
Readiness status (final run only)	reports/tests/automationReadiness/index.html test-results/automationReadiness/TEST-*.xml
Automation checkpoint screenshots	screenshots/milestoneFAutomation/successful-tests/<AUT-F ID>/ (final run) screenshots/milestoneFAutomation/failed-tests/<AUT-F ID>/ (diagnostic)
Gradle problems report	reports/problems/problems-report.html

The root build/reports/problems/problems-report.html came from the 28 September readiness check. The final passing Milestone-F run has its own problems report, with the same three Gradle deprecation warnings. They concern future Gradle compatibility. Use the Milestone-F HTML/XML and exit code for test status. That run passed 6 of 6 on Gradle 8.14.5.

6.3 Evidence in Milestone-F

The evidence copies are stored under Milestone-F/ in the repository. The table below shows where to find them and what they contain.

Path under Milestone-F/	Evidence files
Execution Results/verified-run-20260930-210141-857/	verified-run.log and four XML files show that the readiness test and all six F tests passed, with Gradle exit code 0. Ten browser checkpoint PNGs are under screenshots/milestoneFAutomation/successful-tests/, grouped by AUT-F01 to AUT-F06. Chrome was maximized for this run.
Execution Results/diagnostic-20260930-122320-911/	failure-diagnostic.log and three XML files show six deliberately induced failures and exit code 1. Six failure PNGs are under screenshots/milestoneFAutomation/failed-tests/, one for each AUT-F01 to AUT-F06. This earlier run took place before the Chrome maximization change.
Evidence/	suite-summary.png shows all six F test outcomes. checkout-tests.png shows AUT-F04 and AUT-F06. tested-source-sha256.md lists the tested source checksums and the two workbook hashes carried forward from M-F-A1.

Gradle generates compiled classes and temporary files in build/, so the whole folder was not copied for publication. The complete HTML reports remain under build/timestamped-runs/ in the local checkout. The logs, XML results and browser screenshots listed above have been copied into the matching Milestone-F/Execution Results/ folders. The prepared logs and XML files omit local path prefixes.

Milestone-F source was uncommitted when the final run was captured. Evidence/tested-source-sha256.md lists the tested file hashes alongside base HEAD 0570842b8da0c0f4000b21e590f70a4618c97f71. Both protected workbook hashes matched M-F-A1. I reviewed ten checkpoint screenshots from the passing run against the expected page states.

7 Scope and limitations

The Milestone-E manual cases provided the test basis for Milestone-F. The six automated workflows use their steps, data and expected results. AUT-F03 brings together the related cart cases, while F06 checks the checkout overview and completion results. When the scripts check prices or totals, they compare the displayed values within the same run. These six workflows demonstrate Selenium automation skills within a limited scope. They were not selected through a client-approved automation plan or formal risk ranking.

The tests ran in the maintained Java lane using Chrome, with a fresh browser for each test. The final full run passed all six tests. Its HTML and XML reports show the test status, and ten screenshots show verified checkpoints. A separate diagnostic run deliberately caused six failures to check that failed status and browser images were captured. The deliberately changed expected values were restored before the final passing run. Those diagnostic failures were not SauceDemo defects. The passing evidence establishes the outcome of one full Chrome suite run, but does not establish repeat-run stability or results in other browsers.

Coverage is limited to six SauceDemo workflows using the standard and locked-out accounts. AUT-F04 checks a blank First Name field, and AUT-F05 checks prices sorted from low to high. AUT-F06 checks that the PDF control is present on the completion page. It does not generate a receipt. The other Milestone-E variations remain manual results. Milestone-F uses the inherited repository's Gradle build. Its six SauceDemo tests run in the separate maintainedTest lane with its own Selenium and TestNG setup. The results do not cover the full SauceDemo site or the inherited AutomationPractice suite.

Milestone-F added a dedicated Gradle task and new Java tests and supporting classes within the inherited repository's maintained lane. The tests use the Selenium and TestNG versions established there. They do not run through the older suite or import its reporting and driver helpers. Developing these scripts involved locator work, assertions and failure investigation alongside the Selenium maintenance and enhancement skills used in testing work. The application and test accounts are public demonstration resources. The results do not represent a client sign-off or a release decision.

8 Conclusion

The aim of Milestone-F was to extend the maintained lane with a small set of Selenium workflows linked to Milestone-E and to produce results that could be checked. That technical aim has been met. The final run executed six independent tests once and all passed, including the two-item checkout. Their assertions checked page states and displayed values against Milestone-E's documented outcomes. The separate deliberate-failure run confirmed that a wrong expectation makes the Gradle task fail. Its test report retained the failed assertion, and a browser image was saved before Chrome closed. The evidence review linked the passing result to the tested source. The original Milestone-E workbooks remained unchanged.

My earlier Java Selenium work involved maintaining and enhancing existing scripts. In Milestone-F, I developed six new tests and supporting classes in the established maintained lane. The passing and deliberate-failure runs show the script development and failure analysis carried out within this scope. Designing a new Selenium framework and setting an employer's automation strategy were outside this project. The work reused the lane's tools rather than importing helpers from the older suite. When a browser warning interrupted AUT-F03, I examined the result and retested after adjusting the automation browser setting. The six workflows cover a defined part of SauceDemo. They do not represent full website coverage or a decision about employer automation priorities.